**Biospectra : Vol. 11(1), March, 2016, pp.5-10**
*An International Biannual Refereed Journal of Life Sciences*

***Animal Science***

ISSN : 0973-7057

# Effect of per oral inoculation of bacterial isolates from flacherie disease silkworm on larval weight reduction

**Anusha, H. G. & Bhaskar, R.N.***
*Department of Sericulture UAS, GKVK, Bengaluru-65, Karnataka, India

*Received : 22nd December, 2015; Revised : 27th February, 2016*

**Abstract :** Flacherie is a major disease in silkworm and it is a flaccid condition in which worm suffer from vomiting dysentery. The disease is common during summer and rainy seasons in all the sericultural areas of India and it is caused by different species of bacteria, viruses and their mixed infections. Per oral inoculation of different pathogenic bacteria *viz., Lysinibacillus sphericus*(L), *Alcaligenes faecalis(A), Bacillus subtilis(B)* and NPV (N) revealed significant reduction in larval weight from end of the third instar to subsequent instars(fourth and fifith instar). The administration of *Lysinibacillus sphericus* with other bacterial isolates to third instar larvae of PM×$CSR_2$ recorded maximum of 0.99 g/10 recorded for distilled water administered lot and the same lot noticed minimum of 0.76 g/10(L+A+N) at the end of the third instar and the same trend was found in both fourth and fifth instar batch. The combined effect of NPV and other bacterial isolates also revealed that, the decreased larval weight ranges from 0.77 (N+B+L) - 0.83 (N+A)at the end of third instar, where as 0.82 (N+L+A) - 0.86 (N+A) and 2.76 (N+B+L) - 3.00 (N+A) , 2.89 (N+B+L) - 3.18 (N+A) and 18.65 (N+B+L) - 19.40 (N+A) g/10 at the beginning and end of the fourth and fifth instar over distilled water administered batch(0.99, 1.04 &4.60, 4.68 &24.85 g/10 larvae) respectively. The pathogenicity of the bacterial isolate *Alcaligenes faecalis* was low compared to *Lysinibacillus sphericus*(L), *Bacillus subtilis(B)* and NPV(N).

**Keywords : Bacterial isolates, PM×CSR2, larval weight, B. mori.**

## INTRODUCTION

Silkworm, *Bombyx mori* L. is an important economic Lepidopteron insect and utilized for the commercial production of natural silk fiber Queen of Textile. Its application in all over the world finely contributed to promotion as a powerful laboratory model for the basic and applied research in biology. India occupies second position in raw silk production in the world, its productivity per unit area is low compared to sericulturally advanced countries. Country's raw silk production has reached a new height of 28708 MT with a growth rate of 8.4% over the previous year (Anonymous, 2015).

Silkworm being a poikilothermic, it cannot regulate its temperature and susceptible to various kinds of infectious diseases *viz.,* pebrine, muscardine, flacherie and NPV caused by various pathogenic microorganisms. These microorganisms are not only found in the diseased worms but also scattered in the rearing room and on appliances from the carcasses and feces of diseased caterpillars. These infectious microbes left over become an easy source of secondary contamination and spread of diseases (Gangawar, 2011). Among different diseases the Bacterial Flacherie is considered to be a major disease in silkworm and flaccidity of larva is the major symptom. The disease is very common during summer and rainy seasons in all the sericultural areas of India which accounts to crop loss of 27 – 35% with cocoon yield loss of 11 – 15kg / 100 dfls. It is reported that, flacherie is caused by different species of bacteria, viruses and their mixed infections (Selvakumar *et al.,* 2013).Therefore the study was carried on " Effect of per oral inoculation of bacterial isolates from flacherie diseases silkworm on larval weight reduction".

*Corresponding author :
Phone :
E-mail : rn_bhaskar@rediffmail.com

## MATERIAL AND METHODS

The present study was carried during 2014-16 in the Department of Sericulture, University of Agricultural Sciences, GKVK, Bangaluru-65. The known volume of inoculum administered to silkworm at the rate of 0.25 ml per 25 larvae with $10^7$ dilution to each replication at the beginning (6h after moulting) of the third instar and middle of the instar. The inoculums were administered first singly. Further, bacterial isolates were inoculated in combination of two *viz.,* Bacteria+ virus and virus +bacteria and three organisms *viz.,*bacteria + two bacteria, Virus +two bacteria, bacteria+(virus+bacteria).While inoculating two organisms, first organism was inoculated at the beginning of the instar and the second organism (isolate) was inoculated during middle of the instar whereas in case of three organism administration, first organism is administered at the beginning and the other two organismswere inoculated in the middle of the instar. The bacterial and viral pathogens were identified through molecular characterization and reported as *Lysinibacillus sphericus, Alcaligenes faecalis, Bacillus subtilis and* NPV. The one way anova analysis was used for data analysis.

Treatments detail are given below

**I. *Lysinibacillus sphaericus* with other organisms**

$T_1$-*Lysinibacillus sphericus*

$T_2$- *Lysinibacillus sphericus* +*Alcaligenes faecalis*

$T_3$- *Lysinibacillus sphericus* +*Bacillus subtilis*

$T_4$- *Lysinibacillus sphericus* +NPV

$T_5$- *Lysinibacillus sphericus*+ *Alcaligenes faecalis*+ *Bacillus subtilis*

$T_6$- *Lysinibacillus sphericus*+ *Alcaligenes faecalis*+ NPV

$T_7$- Normal Leaf

$T_8$- Distilled water

II. ***Bacillus subtilis* with other organisms**

$T_1$- *Bacillus subtilis*

$T_2$- *Bacillus subtilis*+ *Lysinibacillus sphericus*

$T_3$- *Bacillus subtilis*+ *Alcaligenes faecalis*

$T_4$- *Bacillus subtilis*+ NPV

$T_5$- *Bacillus subtilis*+ *Lysinibacillus sphericus*+ *Alcaligenes faecalis*

$T_6$- *Bacillus subtilis*+ *Lysinibacillus sphericus*+ NPV

$T_7$- Normal Leaf

$T_8$- Distilled water

**III. *Alcaligenes faecalis* with other organisms**

$T_1$- *Alcaligenes faecalis*

$T_2$-*Alcaligenes faecalis*+ *Lysinibacillus sphericus*

$T_3$- *Alcaligenes faecalis*+ *Bacillus subtilis*

$T_4$- *Alcaligenes faecalis*+ NPV

$T_5$- *Alcaligenes faecalis*+ *Bacillus subtilis*+ *Lysinibacillus sphericus*

$T_6$- *Alcaligenes faecalis*+ *Bacillus subtilis*+ NPV

$T_7$- Normal Leaf

$T_8$- Distilled water

***IV. NPV with* other organisms**

$T_1$- NPV

$T_2$- NPV+ *Lysinibacillus sphericus*

$T_3$- NPV+ *Alcaligenes faecalis*

$T_4$-NPV+ *Bacillus subtilis*

$T_5$- NPV+ *Lysinibacillus sphericus*+ *Alcaligenes faecalis*

$T_6$- NPV+ *Alcaligenes faecalis*+ *Bacillus subtilis*

$T_7$- NPV+ *Bacillus subtilis*+ *Lysinibacillus sphericus*

$T_{8-}$ Normal Leaf

$T_9$- Distilled water

**Breed:** PM×$CSR_2$

## RESULTS AND DISCUSSION

The experimental data, on larval weight of third instar was recorded after per oral inoculation of different groups of bacteria and their combined effect has revealed significant difference both in beginning and end of fourth and fifth instar where as no significant results was noticed at the beginning of the third instar. The data on larval weight inoculated with *Lysinibacillus sphericus*(L), *Alcaligenes faecalis(A), Bacillus subtilis(B)* and NPV (N) revealed significant difference between individual administration and combined administration. Further, administration of *Lysinibacillus sphericus* with other organisms to third instar larvae of PM×$CSR_2$ was recorded maximum of 0.99 g/10 for distilled water administered lot and the same lot noticed minimum of 0.76 g/10(L+A+N) at the end of the third instar. Further, the data revealed that, even in fourth instar of beginning and end, minimum larval weight of 0.80 and 2.71 g/10 (L+A+N) and maximum of 1.04 and 4.6 g/10 was noticed for distilled water administered batch. The trend was same even in beginning (2.90), middle(13.64) and end (18.90) g/10 of fifth instar respectively. It is clearly indicated that, combined effect of (L+A), (L+B), (L+N), (L+A+B) and (L+A+N) revealed significant difference (Table1).

Even in *Bacillus subtilis* administarion to third instar larvae of PM×$CSR_2$ revealed variation in larval weight from

individual administration (0.79 g/10) which was recorded at the end of third instar. Among different bacterial isolate administration, maximum larval weight of 0.99 g/10 was recorded at the end of third instar, where as 1.04 and 4.60 g/10 was recorded during beginning and end of fourth instar. In case of fifth instar, the larval weight of 4.68 , 22.96 & 24.85g/10 was recorded at beginning, midlle and end of the distilled water administered batch and minimum larval weight was recorded when inoculated with B+L+N(3.19,15.44 and 16.00) followed by B+N(3.20,16.54 and 17.80) (Table 2).

The same trend was seen even in *Alcaligenes faecalis* administration to third instar larvae but the interaction effect of two organisms *viz.,* A+L(0.23 & 0.82, 0.89 & 3.41, 3.60& 21.99 g/10 larvae), A+B (0.24 & 0.81, 0.88 & 3.52, 3.70 & 22.0 g/10 larvae) and A+N(0.25 & 0.79, 0.83 & 3.10, 3.14 & 21.10 g/10 ) and three organisms includes A+B+L (0.23 & 0.85, 0.89 & 3.20, 3.35 & 20.10 g/10 larvae) and A+B+N(0.23 & 0.78, 0.80 & 2.47, 3.00 & 18.70) g/10 revealed significantly reduced the larval weight during third, fourth and fifth instar beginning and end of the instar(Table3).

The combined effect of NPV and other bacterial isolates also revealed decreased larval weight ranges from 0.77 (N+B+L) to 0.83 (N+A) at the end of third instar, where as 0.82 (N+L+A) to 0.86 (N+A) and 2.76 (N+B+L) to 3.00 (N+A), 2.89 (N+B+L) to 3.18 (N+A) and 18.65 (N+B+L) to 19.40 (N+A) g/10 at the beginning and end of the fourth and fifth instar over distilled water administered batch(0.99, 1.04 &4.60, 4.68 &24.85 g/10 larvae) respectively**.**

These experimental results are in confirmative with findings of Doreswamy *et al.* (2004) who studied the seasonal incidence of flacherie disease and its influence on silkworm *Bombyx mori*. He reported that, the incidence of flacherie disease was more during summer season followed by winter and rainy season and recorded the least larval weight of 26.45 and 26.43 g/10 larvae during April and May. The highest larval weight was recorded during December (34.88 g/10) and January (34.50 g/10).

The third instar silkworms PM and CSR2 was inoculated with bacteria isolated from the larval surface revealed that, the larval weight recorded at the beginning and end of the instars 0.62, 1.91 and 0.90, 1.28 g/10 larvae which was lower than the healthy batch (0.65, 2.37 and 1.18, 1.24 g/10 larvae). Further, when same batch attain the $4^{th}$ and $5^{th}$ instar recorded less larval weight in the inoculated one compared to healthy worms (Anusha, 2013). Further, laraval weight was significantly less (23.04 g/10 larvae) in inoculation with *Bm*DNV + *Bacillus* sp.which were found on par with *Bm*IIFV+*Bm*DNV, *Bm*IFV+*Bm*DNV+*Bacillus* sp., *Bm*IFV+*Bm*DNV+*S. faecalis, Bm*IFV+*Bm*DNV+ *S. aureus*, *Bm*DNV+ *S. faecalis* and *Bm*DNV+ *S. aureus* in the mean dilutions of Stock, $10^{-2}$, $10^{-4}$, $10^{-8}$, $10^{-16}$, $10^{-32}$. The highest larval weight of (39.00 g/10 larvae) was recorded in untreated and distilled water control (36.00 g/10) (Doreswamy, 2002).

## CONCLUSION

The data concluded that combined administration of NPV with two and three bacterial isolates *viz.,* (NPV+ *Lysinibacillus sphericus,* NPV+ *Alcaligenes faecalis and* NPV+ *Bacillus subtilis*) and three (NPV+ *Lysinibacillus sphericus*+ *Alcaligenes faecalis,* NPV+ *Alcaligenes faecalis*+ *Bacillus subtilis*, and NPV+ *Bacillus subtilis*+ *Lysinibacillus sphericus)* revealed minimum larval weight at the end of the fifth instar followed by *Bacillus subtilis* with other organisms, *Lysinibacillus sphericus* with other organisms and less larval weight was recorded in *Alcaligenes faecalis* with other organisms administered batches and it is reported as less pathogenic to silkworm compared to other organisms.

**Table1: Combined effect of *Lysinibacillus sphaericus* with other organisms on larval weight reduction (g/10) of PM× $CSR_2$ (Third instar inoculated)**

| | Instars | | | | | | |
|---|---|---|---|---|---|---|---|
| | III | | IV | | V | | |
| | Beginning | End | Beginning | End | Beginning | Middle | End |
| L | 0.23 | 0.83 | 0.90 | 3.86 | 3.94 | 19.4 | 21.00 |
| L+A | 0.24 | 0.84 | 0.87 | 3.54 | 3.72 | 17.45 | 21.99 |
| L+B | 0.23 | 0.82 | 0.85 | 3.52 | 3.70 | 18.78 | 20.09 |
| L+N | 0.24 | 0.79 | 0.82 | 3.22 | 3.46 | 17.70 | 19.99 |
| L+A+B | 0.24 | 0.82 | 0.85 | 2.88 | 2.98 | 18.47 | 20.10 |
| L+A+N | 0.24 | 0.76 | 0.80 | 2.71 | 2.90 | 13.64 | 18.90 |
| Cntrol | 0.24 | 0.96 | 0.99 | 4.50 | 4.60 | 22.50 | 24.65 |
| DW | 0.25 | 0.99 | 1.04 | 4.60 | 4.68 | 22.90 | 24.85 |
| F test | NS | ** | ** | ** | ** | ** | ** |
| S.Em ± | 0.01 | 0.03 | 0.04 | 0.06 | 0.05 | 0.53 | 0.20 |
| CD at 5% | 0.04 | 0.06 | 0.08 | 0.14 | 0.12 | 1.13 | 0.43 |

*L-Lysinibacillus sphaericus, A-Alcaligenes faecalis, B-Bacillus subtilis and N*-NPV
** Significant at 1 %, NS non significant

**Table 2: Combined effect of *Bacillus subtilis* with other organisms on larval weight (g/10) of PM×$CSR_2$ (Third instar inoculated)**

| | Instars | | | | | | |
|---|---|---|---|---|---|---|---|
| | III | | IV | | V | | |
| | Beginning | End | Beginning | End | Beginning | Middle | End |
| B | 0.24 | 0.79 | 0.85 | 3.84 | 3.92 | 18.70 | 20.10 |
| B+L | 0.24 | 0.84 | 0.89 | 3.30 | 3.46 | 19.98 | 20.18 |
| B+A | 0.22 | 0.80 | 0.87 | 3.40 | 3.58 | 18.62 | 19.98 |
| B+N | 0.23 | 0.77 | 0.80 | 3.07 | 3.20 | 16.54 | 17.80 |
| B+L+A | 0.24 | 0.81 | 0.87 | 3.96 | 4.00 | 16.85 | 17.89 |
| B+L+N | 0.22 | 0.78 | 0.83 | 3.02 | 3.19 | 15.44 | 16.00 |
| Cntrol | 0.24 | 0.96 | 0.99 | 4.50 | 4.60 | 22.50 | 24.65 |
| DW | 0.25 | 0.99 | 1.04 | 4.60 | 4.68 | 22.96 | 24.85 |
| F test | NS | ** | ** | ** | ** | ** | ** |
| S.Em ± | 0.01 | 0.02 | 0.03 | 0.12 | 0.29 | 0.55 | 0.39 |
| CD at 5% | 0.03 | 0.05 | 0.06 | 0.26 | 0.63 | 1.18 | 0.82 |

*B-Bacillus subtilis, L-Lysinibacillus sphaericus, A-Alcaligenes faecalis, and N*-NPV
** Significant at 1 %, NS non significant

**Table 3: Combined effect of *Alcaligenes faecalis with* other organisms on larval weight (g/10) of PM×$CSR_2$ (Third instar inoculated)**

| | Instars | | | | | | |
|---|---|---|---|---|---|---|---|
| | III | | IV | | V | | |
| | Beginning | End | Beginning | End | Beginning | Middle | End |
| A | 0.24 | 0.81 | 0.87 | 3.04 | 3.24 | 18.27 | 20.08 |
| A+L | 0.23 | 0.82 | 0.89 | 3.41 | 3.60 | 20.02 | 21.99 |
| A+B | 0.24 | 0.81 | 0.88 | 3.52 | 3.70 | 20.00 | 22.00 |
| A+N | 0.25 | 0.79 | 0.83 | 3.10 | 3.14 | 18.67 | 21.10 |
| A+B+L | 0.23 | 0.85 | 0.89 | 3.20 | 3.35 | 19.40 | 20.10 |
| A+B+N | 0.23 | 0.78 | 0.80 | 2.47 | 3.00 | 15.28 | 18.70 |
| Cntrol | 0.24 | 0.96 | 0.99 | 4.50 | 4.60 | 22.50 | 24.65 |
| DW | 0.25 | 0.99 | 1.04 | 4.60 | 4.68 | 22.90 | 24.85 |
| F test | NS | ** | ** | ** | ** | ** | ** |
| S.Em ± | 0.01 | 0.03 | 0.02 | 0.06 | 0.15 | 0.27 | 0.47 |
| CD at 5% | 0.03 | 0.08 | 0.04 | 0.14 | 0.32 | 0.58 | 0.99 |

*A-Alcaligenes faecalis, L-Lysinibacillus sphaericus, B-Bacillus subtilis and N*-NPV

** Significant at 1 %, NS non significant

**Table 4: Combined effect of NPV *with* other organisms on larval weight (g/10) of PM×$CSR_2$ (Third instar inoculated)**

| | Instars | | | | | | |
|---|---|---|---|---|---|---|---|
| | III | | IV | | V | | |
| | Beginning | End | Beginning | End | Beginning | Middle | End |
| N | 0.23 | 0.77 | 0.81 | 2.80 | 2.89 | 19.10 | 19.30 |
| N+L | 0.22 | 0.79 | 0.82 | 2.90 | 2.98 | 17.30 | 19.00 |
| N+A | 0.24 | 0.83 | 0.86 | 3.00 | 3.18 | 18.98 | 19.40 |
| N+B | 0.25 | 0.80 | 0.84 | 2.89 | 3.00 | 17.86 | 18.89 |
| N+L+A | 0.24 | 0.79 | 0.82 | 2.84 | 2.98 | 16.80 | 18.70 |
| N+A+B | 0.23 | 0.81 | 0.83 | 2.82 | 3.00 | 16.73 | 18.73 |
| N+B+L | 0.25 | 0.77 | 0.82 | 2.76 | 2.89 | 16.99 | 18.65 |
| Cntrol | 0.24 | 0.96 | 0.99 | 4.50 | 4.60 | 22.50 | 24.65 |
| DW | 0.25 | 0.99 | 1.04 | 4.60 | 4.68 | 22.90 | 24.85 |
| F test | NS | ** | ** | ** | ** | ** | ** |
| S.Em ± | 0.02 | 0.03 | 0.04 | 0.07 | 0.05 | 0.14 | 0.27 |
| CD at 5% | 0.05 | 0.07 | 0.08 | 0.15 | 0.11 | 0.30 | 0.57 |

*N*-NPV *and L-Lysinibacillus sphaericus, A-Alcaligenes faecalis, B-Bacillus subtilis*

** Significant at 1 %, NS non significant

## REFERENCES


1. **Anonymous, 2015,** Annual report, *Cetral silk Board,Madiwala,Bengaluru*-68,P:122
2. **Anusha, H. G.,2013,** studies on farmers practices of silkworm (*Bombyx mori* l.) rearing and *in vivo Bacillus* infection in zone – 5 (Kolar AND Chikkaballapur Districts). *M.Sc thesis,UAS, GKVK, Bengaluru-65*, P-121.
3. **Doreswamy, C., 2002,** Etiology and epizootiology of late larval falcherie of silkworm bombyx mori L. *Ph.D thesis, UAS, GKVK, Bengaluru-65* p- 271.
4. **Doreswamy, C., Govindhan, R., Devaiah, M. C. and Muniswamappa, M. V. 2004.** Laboratory studies on seasonal incidence of late larval flacherie of silkworm, *Bombyx mori. Bull.Ind.Acad.Seri.,* **9**(2):20-24.
5. **Gangawar, S. K., 2012,** Seasonal response of two (Silkworm, *Bombyx mori* L.) Bivoltine hybrids with comparative performance shoot vs. shelf rearing in Uttar Pradesh climatic conditions, *Bulletin of environment, Pharmacology and life Sciences,* **1**(8): 14-17.

6. **Selvakumar, T, 2013,** Prevalence of flacherie diseases and pathogenicity of isolated pathogence in silkworm *Bombyx mori* under different environmental conditions. *Agri.Sci.digest.,***33**(4):253-258.

* * *